Troubleshooting and Resolving ASIC Memory Parity Errors and Line Card Crashes on Huawei Access Node YEL-ACC-22

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5395

Description

The Huawei Access Node YEL-ACC-22, situated at the Yelahanka Sub-station, is equipped with the Huawei NetEngine 8000 series, a high-performance access node designed to provide efficient packet processing and forwarding capabilities. The occurrence of ASIC memory parity errors and line card crashes in this specific role and vendor equipment can be attributed to the inherent complexities of the ASIC memory and the line card's ability to handle high-traffic volumes. The ASIC memory, being a critical component, is responsible for storing and processing packet forwarding information. Any errors or malfunctions in the ASIC memory can lead to incorrect packet forwarding, resulting in line card crashes. Furthermore, the Huawei NetEngine 8000 series utilizes advanced ASIC designs to enhance packet processing efficiency. However, the high-speed and high-density nature of these ASICs can lead to increased susceptibility to memory parity errors and line card crashes. Environmental factors, such as temperature fluctuations, power supply instability, or electrical noise, can also contribute to the occurrence of these faults. As a result, it is essential to perform thorough diagnostic and resolution procedures to identify and rectify the root cause of the issue.

Pre-requisite Checks

- Verify that the Yelahanka Sub-station has power and is accessible by remote hands.
- Check the status of the redundant link to ensure it is operational.
- Confirm that the Huawei NetEngine 8000 series node is running the latest software version.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors due to Manufacturing Defects or Environmental Factors	High
Line Card Crashes due to High-Traffic Volumes or Software Issues	Medium
Power Supply Redundancy Loss due to Hardware Failure or Configuration Issues	Low

Diagnostic Commands & Expected Outputs

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Command: display device asic memory info, Expected Output: ASIC memory information, including memory size, type, and error counts., Description: This command retrieves information about the ASIC memory, including its size, type, and error counts, allowing for identification of potential memory parity errors.
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Command: display linecard crash info, Expected Output: Line card crash information, including reason, time, and location., Description: This command retrieves information about line card crashes, including the reason, time, and location, allowing for identification of potential software or hardware issues.

Command: display power supply status, Expected Output: Power supply status, including operation, voltage, and current., Description: This command retrieves information about the power supply status, including operation, voltage, and current, allowing for verification of power supply redundancy.
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Resolution Steps

Step 1: Step: Replace Faulty ASIC Memory, Command: replace device asic memory, Description: This step involves replacing the faulty ASIC memory module to resolve memory parity errors. Ensure that the replacement module is compatible with the node's specifications and is properly installed.

Step 2: Step: Upgrade Software to Latest Version, Command: upgrade system software, Description: This step involves upgrading the system software to the latest version to resolve software issues. Ensure that the upgrade is performed during maintenance windows and that the new software version is compatible with the node's hardware configuration.

Step 3: Step: Verify Power Supply Redundancy, Command: verify power supply redundancy, Description: This step involves verifying the power supply redundancy to ensure that it is operational and functioning correctly. Check the power supply status and ensure that it is properly configured.

Post-Check Verification

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Command: display device asic memory info, Expected Output: ASIC memory information, including memory size, type, and error counts., Description: This command retrieves information about the ASIC memory, including its size, type, and error counts, allowing for verification of resolution.

Command: display linecard crash info, Expected Output: Line card crash information, including reason, time, and location., Description: This command retrieves information about line card crashes, including the reason, time, and location, allowing for verification of resolution.

Command: display power supply status, Expected Output: Power supply status, including operation, voltage, and current., Description: This command retrieves information about the power supply status, including operation, voltage, and current, allowing for verification of power supply redundancy.
```

Roll-back Procedures

- Step: Rollback Software Upgrade, Command: rollback system software, Description: This step involves rolling back the software upgrade to the previous version in case the new software version causes issues.

- Step: Restore Original ASIC Memory, Command: restore device asic memory, Description: This step involves restoring the original ASIC memory module in case the replacement module causes issues.

- Step: Verify Power Supply Redundancy, Command: verify power supply redundancy, Description: This step involves verifying the power supply redundancy to ensure that it is operational and functioning correctly.